AUTOMATED TRAFFIC VIOLATION DETECTION SYSTEM

PROJECT REPORT

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BONAFIDE CERTIFICATE

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TABLE OF CONTENTS

TABLE OF CONTENTS

CHAPTER NO	TITLE	PAGE NO
	ABSTRACT	
	LIST OF ABBREVATIONS	
	LIST OF FIGURES	
1	INTRODUCTION	1
2	LITERATURE SURVEY	4
3	TECHNOLOGY STACK	14
4	IMPLEMENTATION OF TRAFFIC VIOLATION DETECTION SYSTEM	18
	4.1 EXISTING SYSTEM	
	4.2 DRAWBACKS	
	4.3 PROPOSED SYSTEM	
	4.4 MERITS	
	4.5 ALGORITHM	
	4.6 FLOW CHART	
5	RESULT AND DISCUSSION	31
6	CONCLUSION AND FUTURE ENHANCEMENT	33
7	APPENDICES	
	FRONTEND SOURCE CODE	35
	BACKEND SOURCE CODE	37
	OUTPUT	40
8	REFERENCES	44

ABSTRACT

Road accidents have become a major global concern, with a significant number of fatalities occurring due to the negligence of traffic safety measures. Among these, the failure of two-wheeler riders to wear helmets is one of the leading causes of severe head injuries and deaths. Traditional methods of monitoring such violations rely heavily on manual observation, which is inefficient, time-consuming, and prone to human error. Therefore, there is a growing need for an automated system that can ensure effective monitoring and enforcement of traffic rules.

This project presents an AI-based traffic surveillance system that focuses on detecting motorcyclists who are not wearing helmets. The system utilizes advanced computer vision techniques to analyze video input captured from CCTV cameras or recorded traffic footage. OpenCV is used for processing video frames, enabling the system to extract and analyze images in real time. This forms the foundation for further object detection and violation identification.

A YOLO (You Only Look Once) based deep learning model is employed to detect motorcycles and identify whether the rider is wearing a helmet. YOLO is known for its high speed and accuracy in real-time object detection, making it suitable for traffic monitoring applications. When a helmet violation is detected, the system highlights the rider and captures the corresponding frame, ensuring that the violation is properly recorded for further processing.

To identify the violating vehicle, the system extracts the number plate region from the captured frame. The processed image is then passed to Tesseract OCR, which extracts the vehicle registration number. This automated number plate recognition helps in accurately identifying violators without manual intervention.

LIST OF ABBREVIATIONS

UI – User **I**nterface

UX – User **E**xperience

YOLO – You **O**nly **L**ook **O**nce

ANPR – Automated **N**umber **P**late **R**ecognition

OCR – Optical **C**haracter **R**ecognition

Git – Version **C**ontrol **S**ystem

GitHub – Code **H**osting **P**latform

JWT – JSON Web **T**oken (Authentication)

LIST OF FIGURES

F. NO	TITLE	PAGE NO
4.1	System Architecture of Traffic Violation Detection System	21
4.2	Flowchart of Traffic Violation Detection System	30
4.3	Dashboard	40
4.4	Live Camera Feed Page	40
4.5	Violation Page	41
4.6	Plate Records Page	42
4.7	Analytics Page	42
4.8	Project Display & Search Filtering	43

CHAPTER 1

INTRODUCTION

Road safety has become a critical issue worldwide due to the rapid increase in the number of vehicles and traffic density. Among various types of road users, two-wheeler riders are more vulnerable to accidents and injuries. One of the primary reasons for severe injuries and fatalities is the failure to wear helmets while riding. Despite strict traffic rules and awareness campaigns, many riders neglect helmet usage, leading to serious consequences. Ensuring compliance with helmet laws through manual monitoring is difficult, especially in busy urban environments.

Traditional traffic monitoring systems rely heavily on human intervention, where traffic police or surveillance personnel manually observe violations through direct monitoring or CCTV footage. This approach is not only time-consuming but also prone to human error, fatigue, and inefficiency. In high-traffic areas, it becomes almost impossible to monitor every vehicle continuously. As a result, many violations go unnoticed, reducing the effectiveness of traffic law enforcement and compromising road safety.

This project proposes an AI-based traffic surveillance system specifically designed to detect motorcyclists who are not wearing helmets. The system processes video input using OpenCV and applies a YOLO-based model to detect motorcycles and helmet usage. When a violation is identified, the system captures the frame and extracts the vehicle number plate using OCR techniques. The detected information is then displayed on a dashboard for monitoring. By focusing on helmet detection for two-wheelers, the system aims to enhance road safety and provide a practical solution for modern traffic management.

1.1 Objective of the Project:

The primary objective of this project is to develop an AI-based system capable of automatically detecting motorcyclists who are not wearing helmets. The system aims to utilize computer vision techniques to process video input from CCTV cameras and analyze rider behavior in real time. By implementing a YOLO-based deep learning model, the system can accurately identify motorcycles and determine helmet usage without human intervention. This helps in reducing manual monitoring efforts and ensures continuous surveillance in high-traffic areas.

Another important objective of the project is to identify and record the details of violating vehicles. Once a helmet violation is detected, the system extracts the number plate region from the captured frame and uses Tesseract OCR to convert it into readable text. The extracted information is then displayed on a dashboard for monitoring and further analysis. Overall, the project aims to improve traffic law enforcement, enhance road safety, and demonstrate the effective use of artificial intelligence in real-world applications.

1.2 Overview of the Project:

The AI-Based Traffic Surveillance System is designed to detect motorcyclists who are not wearing helmets using computer vision and deep learning techniques. The system takes video input from CCTV cameras or recorded footage and processes it using OpenCV to extract frames. A YOLO-based model is used to detect motorcycles and identify helmet usage in real time. When a violation is detected, the system captures the frame and highlights the rider. The number plate region is then extracted and processed using Tesseract OCR to obtain the vehicle registration number. The detected information is sent to a Flask backend and displayed on a React dashboard for monitoring. This project focuses only on helmet detection for two-wheelers and provides an automated solution to improve traffic safety and enforcement.

1.3 Scope of the Project:

The scope of this project focuses on developing an automated system for detecting helmet violations among two-wheeler riders using computer vision and deep learning techniques. The system processes video input from CCTV cameras or recorded footage to identify motorcycles and determine whether riders are wearing helmets. It further extends to extracting number plate information using OCR, enabling the identification of violators. This project demonstrates the practical application of AI in real-time traffic monitoring and enforcement.

In addition, the system can be expanded for use in smart city applications and large-scale traffic management systems. Future enhancements may include detection of other traffic violations such as over-speeding, signal jumping, and integration with automated fine generation systems. The current model provides a strong foundation for building more advanced intelligent transportation systems that improve road safety and reduce human intervention.

1.4 Significance of the Project:

This project is significant as it addresses one of the major causes of road accidents—failure to wear helmets by two-wheeler riders. By automating the detection of helmet violations using artificial intelligence and computer vision, the system reduces dependence on manual monitoring and improves the efficiency of traffic law enforcement. It ensures continuous surveillance, minimizes human error, and helps authorities identify violators quickly and accurate.

Platform Furthermore, the project demonstrates the practical use of modern technologies such as OpenCV, YOLO, and Tesseract OCR in solving real-world problems. It contributes to the development of smart traffic management systems and promotes safer road practices. The system can be scaled and enhanced for broader applications, making it valuable for solutions and smart city initiatives.

CHAPTER 2

LITERATURE SURVEY

2.1 Helmet Detection

Helmet detection has become an important research area in traffic safety systems, especially for two-wheeler riders. Early approaches relied on traditional image processing techniques such as edge detection, color segmentation, and shape analysis to identify helmets. However, these methods were limited in accuracy and struggled in complex environments with varying lighting conditions and occlusions.

With the advancement of deep learning, modern approaches use object detection models such as YOLO (You Only Look Once) to improve performance. YOLO-based models are capable of detecting motorcycles and identifying helmet usage in real time with high accuracy. Several studies have demonstrated that deep learning models outperform traditional methods by providing faster processing and better adaptability to real-world scenarios. This makes them highly suitable for traffic surveillance systems.

Recent research focuses on optimizing helmet detection models for real-time applications using CCTV footage. These systems are designed to handle multiple vehicles simultaneously and detect violations efficiently. The integration of deep learning techniques has significantly improved detection accuracy and reduced false positives, making automated helmet detection a reliable solution for traffic monitoring.

2.2 Number Plate Recognition:

Number plate recognition is a crucial component of automated traffic surveillance systems, as it enables identification of violating vehicles. Earlier methods used basic image processing techniques such as contour detection and template matching to locate number plates. However, these methods were not robust enough to handle variations in lighting, angle, and image quality..Modern systems use a combination of deep learning models and Optical Character Recognition (OCR) for improved accuracy. Object detection models are used to locate the number plate region, while OCR tools like Tesseract are used to extract text from the image. Preprocessing techniques such as grayscale conversion, noise reduction, and thresholding are applied to enhance the image before text extraction, improving recognition accuracy.

Recent studies have shown that integrating deep learning with OCR provides efficient and reliable number plate recognition in real-time applications. These systems can accurately detect and read vehicle registration numbers from video feeds, even in challenging conditions. This makes them an essential component of intelligent traffic management systems and supports automated enforcement of traffic rules.

Recent advancements in number plate recognition systems have focused on improving accuracy under challenging real-world conditions such as low lighting, motion blur, and varying camera angles. Deep learning-based approaches, particularly those combining convolutional neural networks with OCR, have shown significant improvements in detecting and recognizing number plates from dynamic video streams. These systems are capable of handling different plate formats and fonts, making them more adaptable to diverse environments. As a result, modern number plate recognition techniques play a vital role in automated traffic enforcement by ensuring reliable identification of vehicles involved in violations.

2.3 Cloud-Based Storage and Scalability:

Cloud-based storage plays an important role in enhancing the efficiency and scalability of the proposed traffic surveillance system. In this project, violation data such as captured images, number plate details, and timestamps can be stored in cloud platforms instead of local storage. This allows secure data management, easy access, and prevents data loss. Cloud storage also enables authorities to access violation records from any location, improving monitoring and coordination.

In terms of scalability, cloud integration allows the system to handle large volumes of data generated from multiple CCTV cameras across different locations. As traffic data increases, cloud infrastructure can be easily scaled without affecting system performance. This makes the system suitable for real-world deployment in smart city environments, where continuous monitoring and large-scale data handling are required.

Cloud-based storage also enhances the flexibility and reliability of the proposed system by enabling centralized data management and real-time synchronization. All detected violations, including images and extracted number plate information, can be securely stored and organized in cloud databases, allowing easy retrieval and analysis. This approach reduces dependency on local systems and ensures data availability even in case of hardware failures. Additionally, cloud platforms support integration with advanced analytics tools, which can be used to generate reports, identify traffic patterns, and improve decision-making. The scalable nature of cloud services ensures that as the number of cameras and data volume increases, the system can expand seamlessly without requiring major infrastructure changes, making it highly suitable for future smart traffic management applications.

2.4 User Interface and User Experience Design (UI/UX):

The user interface of the proposed system is designed to be simple, intuitive, and user-friendly, ensuring that traffic authorities can easily monitor violations without technical complexity. A React-based dashboard is used to display real-time data, including live video feed, detected violations, and extracted number plate information. The layout is structured with clear navigation elements such as sidebar menus and organized sections, allowing users to quickly access different features like live monitoring, violation records, and analytics.

The user experience is enhanced by providing clear visual indicators and interactive elements. Detected violations are highlighted with bounding boxes, and status badges are used to indicate whether a rider is wearing a helmet or not. The system also displays important details such as timestamp, location, and vehicle number, making it easier for users to understand and respond to violations. Smooth transitions, responsive design, and consistent color schemes further improve usability across different devices.

In addition, the system includes features such as search, filtering, and data organization to improve efficiency. Users can easily find specific violation records based on number plate or time, reducing the effort required for manual tracking. The interface also supports dark and light modes, enhancing accessibility and comfort during prolonged usage. Overall, the UI/UX design focuses on providing a seamless and efficient experience, ensuring that the system is both functional and easy to operate. The UI/UX design also focuses on responsiveness, ensuring smooth performance across different screen sizes and devices. Clear typography and icon usage improve readability and user interaction. This enhances overall usability and makes the system more efficient for real-time monitoring.

2.5 Secure Authentication and Data Integrity:

Secure authentication is essential to ensure that only authorized users can access the traffic surveillance system. In this project, user access can be controlled through login credentials, where each user is required to provide valid authentication details before accessing the dashboard. This helps prevent unauthorized access to sensitive data such as violation records and number plate information. Implementing authentication mechanisms enhances system security and ensures that only responsible authorities can monitor and manage the system.

Data integrity is another critical aspect of the system, as it ensures that the collected information remains accurate and unaltered during processing and storage. The system maintains the integrity of data by securely transmitting information between the frontend and backend using APIs. Proper validation and error handling mechanisms are applied to ensure that only correct and verified data is stored. This helps in maintaining reliable records of traffic violations, which are essential for enforcement and analysis.

Additionally, security measures can be extended by using encryption techniques and secure communication protocols to protect sensitive data. Storing data in a secure environment, whether locally or on cloud platforms, reduces the risk of data loss or tampering. Regular monitoring and controlled access further enhance the reliability of the system. Overall, secure authentication and data integrity ensure that the system operates safely, maintains trust, and supports effective traffic management. The system can also implement role-based access control to assign different permissions to users such as administrators and operators. Logging mechanisms can be used to track user activities and detect any unauthorized actions. Regular backups ensure data recovery in case of system failures.

2.6 Search and Filtering Process:

The searching and filtering process in the proposed system plays a vital role in efficiently managing and accessing large volumes of traffic violation data. As the system continuously detects helmet violations and extracts number plate information, a significant amount of data is generated and stored. To handle this effectively, the system provides a search functionality that allows users to quickly locate specific records based on key parameters such as vehicle number, date, or time. This reduces manual effort and enables faster retrieval of relevant information.

The filtering mechanism further enhances usability by allowing users to narrow down results based on specific criteria. For example, users can filter violations by date range, location, or type of violation, making it easier to analyze particular datasets. This is especially useful for traffic authorities who need to review violations during a specific period or at a particular location. By applying filters, the system presents only the required data, improving clarity and reducing information overload.

In addition, the integration of search and filtering features improves overall system efficiency and user experience. The interface is designed to provide quick response times, ensuring that results are displayed instantly without delays. These features also support better data analysis and reporting, helping authorities identify patterns and make informed decisions. Overall, the searching and filtering process ensures that the system remains organized, accessible, and effective in managing traffic violation data. The system can also support advanced search options such as partial number plate matching and keyword-based queries. Sorting features can be included to arrange results based on date, time, or violation type.These enhancements further improve data accessibility and system usability.

2.7 Documentation Management in Digital Platforms:

Documentation management plays a crucial role in the successful development and maintenance of the AI-based traffic surveillance system. It involves systematically organizing all project-related information, including design details, system architecture, source code explanations, and user manuals. Proper documentation ensures that every component of the system is clearly defined and easily understandable, which helps developers and users to work efficiently. It also serves as a reference for understanding the workflow, technologies used, and overall functionality of the project.

In this project, documentation includes technical details such as the implementation of OpenCV for video processing, YOLO for object detection, and Tesseract OCR for number plate recognition. It also covers the integration of the Flask backend with the React frontend, explaining how data flows between different components of the system. Additionally, user documentation is provided to guide operators on how to use the dashboard, perform searches, and interpret results. This makes the system more accessible and easier to operate, even for users with limited technical knowledge.

Effective documentation management also supports future enhancements and system scalability. By maintaining clear and updated records, developers can easily modify or extend the system, such as adding new features or improving model accuracy. Documentation also helps in debugging, testing, and troubleshooting by providing insights into system behavior and processes. Overall, proper documentation ensures better project organization, improves communication among team members, and enhances the long-term reliability and usability of the system.

2.8 Importance of Traffic Rules Knowledge:

Knowledge of traffic rules is essential for ensuring road safety and preventing accidents. Traffic regulations are designed to guide the behavior of road users, including drivers, riders, and pedestrians, in order to maintain order and reduce risks on the road. When individuals are aware of rules such as wearing helmets, following speed limits, and obeying traffic signals, they are more likely to act responsibly. This awareness helps in minimizing accidents and protecting lives, especially in the case of two-wheeler riders who are more vulnerable to injuries.

Understanding traffic rules also plays a significant role in improving discipline and efficiency in transportation systems. When road users follow established regulations, traffic flow becomes smoother and more organized, reducing congestion and delays. It also enables authorities to manage traffic effectively and implement enforcement measures more efficiently. Lack of awareness or negligence of traffic rules can lead to chaos, increased violations, and difficulty in maintaining control over road conditions, especially in densely populated urban areas.

In addition, traffic rules knowledge supports the effective implementation of intelligent systems like the proposed AI-based traffic surveillance system. When people are aware that violations such as not wearing helmets can be automatically detected and recorded, it encourages them to follow the rules more strictly. This not only enhances compliance but also strengthens the impact of automated monitoring systems. Overall, awareness and understanding of traffic rules are fundamental to building safer roads, reducing accidents, and promoting responsible driving behavior. Public awareness campaigns and education programs can further improve understanding of traffic rules among road users. This leads to better compliance and contributes to a safer and more disciplined traffic environment.

2.9 Comparative Studies of Traffic Violation:

Comparative studies of traffic violation detection systems highlight the evolution from traditional manual methods to modern automated solutions. In earlier systems, traffic violations such as riding without helmets were identified through direct observation by traffic police or by reviewing CCTV footage manually. These approaches were time-consuming, prone to human error, and limited in their ability to handle large volumes of traffic data. In contrast, modern systems use computer vision and deep learning techniques to automatically detect violations in real time, significantly improving efficiency and accuracy.

When comparing different automated approaches, deep learning models such as YOLO have proven to be highly effective for real-time object detection. Unlike traditional image processing methods that rely on predefined features, YOLO can detect multiple objects in a single frame with high speed and precision. This makes it suitable for identifying motorcycles and detecting helmet usage in dynamic traffic conditions. Additionally, the integration of OCR technologies such as Tesseract enables automatic number plate recognition, further enhancing the capability of these systems to identify and track violators.

The proposed system combines the advantages of both object detection and OCR to provide a more comprehensive traffic monitoring solution. Compared to other systems, it offers real-time detection, automated identification, and user-friendly dashboard visualization. While some advanced systems may include additional features such as multi-violation detection or cloud-based analytics, this project focuses specifically on helmet violation detection for two-wheelers. Overall, comparative studies show that AI-based systems are more reliable, scalable, and efficient than traditional methods, making them ideal for modern traffic management.

2.10 Summary of Literature Findings:

The literature survey highlights the significant advancements made in the field of traffic monitoring through the use of computer vision and artificial intelligence. Earlier systems relied on manual observation and basic image processing techniques, which were limited in accuracy and efficiency. With the introduction of tools like OpenCV, researchers were able to automate certain aspects of traffic analysis, such as vehicle detection and motion tracking. However, these traditional approaches faced challenges in handling real-time scenarios and complex environments, emphasizing the need for more advanced solutions.

Recent studies have demonstrated the effectiveness of deep learning models, particularly YOLO, in improving object detection performance. These models are capable of detecting vehicles and identifying rider behavior with high speed and accuracy, making them suitable for real-time traffic surveillance systems. In addition, the integration of Optical Character Recognition (OCR) techniques, such as Tesseract, has enabled automatic extraction of number plate information. The combination of deep learning and OCR has significantly enhanced the ability to detect violations and identify vehicles without human intervention.

Overall, the literature survey indicates that modern traffic monitoring systems are moving towards fully automated, intelligent solutions that provide real-time detection and analysis. The integration of computer vision, deep learning, and OCR technologies has improved accuracy, efficiency, and scalability in traffic surveillance. The proposed project builds upon these advancements by focusing specifically on helmet violation detection for two-wheelers and incorporating number plate recognition, offering a practical and effective solution for improving road safety and traffic management.

CHAPTER 3

TECHNOLOGY STACK

FRONTEND:

- HTML
- CSS
- JAVASCRIPT (DYNAMIC UI, FORM HANDLING)
- MODALS FOR LOGIN / SIGNUP (SCRIPT.JS)

BACKEND:

- NODE.JS (EXPRESS.JS)
- MONGODB (NOSQL DATABASE)
- MONGOOSE ORM
- JWT AUTHENTICATION
- BCRYPT / AES ENCRYPTION
- RESTFUL APIS

ADDITIONAL TECHNOLOGIES:

- FILE UPLOAD HANDLING
- PROJECT FILTERING & SEARCH ALGORITHMS
- DYNAMIC RENDERING OF PROJECT CARDS (PROJECTS.HTML)
- CLOUD / LOCAL HOSTING SUPPORT

SYSTEM SPECIFICATION:

HARDWARE SPECIFICATIONS:

- 8 GB RAM (Minimum)
- Intel / AMD Processor (i5 or above recommended)
- 256 GB Hard Disk (SSD preferred)
- Stable Internet Connection
- Monitor with minimum 1366×768 resolution
- Minimum 2 GB Graphics Memory (for better visualization and model performance)
- Dedicated GPU (Optional, NVIDIA recommended for faster AI processing)

SOFTWARE SPECIFICATIONS:

- **Operating System:** Windows / macOS / Linux
- **Languages Used:** Python, JavaScript, HTML, CSS
- **Frameworks:** Flask (Backend), React.js (Frontend)
- **Libraries / Tools:** OpenCV, YOLO (Ultralytics), Tesseract OCR, NumPy, Axios
- **OCR Engine:** Tesseract (installed separately)
- **Model Framework:** Ultralytics YOLOv8
- **Tools / IDEs:** VS Code, MongoDB Compass, Chrome Browser
- **Version Control:** Git, GitHub

3.1 Frontend:

The frontend of the proposed system is developed using React.js, providing a dynamic and interactive user interface for monitoring traffic violations. It is designed to display real-time information such as live camera feed, detected helmet violations, and extracted number plate details. The interface includes organized sections like dashboard, live monitoring, and violation records, allowing users to easily navigate and access relevant data. The use of modern web technologies ensures a responsive design that works efficiently across different devices.

In addition, the frontend enhances user experience through clear visual elements such as bounding boxes for detected violations, status indicators, and structured data tables. It communicates with the backend using APIs to fetch and display updated information without delays. Features like search, filtering, and real-time updates make the system more efficient and user-friendly. Overall, the frontend plays a crucial role in presenting processed data in a meaningful and accessible manner, enabling effective monitoring and decision-making.

3.2 Backend:

The backend of the proposed system is developed using Flask, which handles data processing and communication between different components of the system. It receives video frames and detection results, processes them using OpenCV, YOLO, and Tesseract OCR, and extracts relevant information such as helmet violations and number plate details. The backend ensures smooth execution of detection tasks and manages the overall workflow of the system.

In addition, the backend provides API endpoints that allow the frontend to retrieve processed data efficiently. It also handles tasks such as data validation, error handling, and optional storage of violation records. By acting as the core processing unit, the backend ensures accurate detection, reliable data flow, and seamless integration with the frontend dashboard.

3.3 Hardware Specifications:

The hardware requirements of the proposed system are designed to support efficient video processing and real-time detection. A system with a minimum of an Intel or AMD processor (i5 or above) and at least 8 GB RAM is required to handle the computational load of OpenCV and deep learning models like YOLO. Adequate storage, preferably a 256 GB SSD, is necessary to install software, store datasets, and save captured violation images. These specifications ensure smooth performance during continuous video analysis.

In addition, input and output devices such as a CCTV camera or webcam are required for capturing video data, while a monitor with sufficient resolution is needed for displaying the dashboard and detection results. A stable internet connection is useful for downloading libraries, pretrained models, and updates. Optional hardware such as a dedicated GPU can significantly improve processing speed and model performance, making the system more efficient for real-time applications.

3.4 Software Specifications:

The software requirements of the proposed system include an operating system such as Windows or Linux, along with programming languages like Python for backend development and JavaScript for frontend development. Frameworks such as Flask and React.js are used to build the backend and user interface, ensuring smooth communication and efficient data handling.

In addition, the system uses libraries and tools such as OpenCV for image processing, YOLO for object detection, and Tesseract OCR for number plate recognition. Development tools like Visual Studio Code and web browsers such as Google Chrome are used for coding, testing, and debugging the application.

CHAPTER 4

IMPLEMENTATION OF TRAFFIC VIOLATION DETECTION SYSTEM

4.1 EXISTING SYSTEM:

The existing traffic monitoring system primarily relies on manual supervision by traffic police and basic surveillance methods. In many locations, officers are stationed at signals or checkpoints to observe whether riders are following traffic rules such as wearing helmets. This approach requires constant attention and physical presence, making it difficult to manage in areas with heavy traffic. As the number of vehicles increases, it becomes nearly impossible for authorities to monitor every rider effectively.

In addition to manual monitoring, CCTV cameras are widely installed at roads and intersections to record traffic activities. However, these systems mainly function as passive monitoring tools that capture video footage without any automated analysis. The recorded videos are often reviewed later by authorities, which consumes time and delays action against violators. This lack of real-time detection reduces the effectiveness of enforcement and limits the system's ability to prevent violations immediately.

Another major limitation of the existing system is the absence of automation in detecting traffic violations. There is no intelligent mechanism to automatically identify whether a motorcyclist is wearing a helmet. As a result, authorities depend entirely on human observation, which can lead to inconsistencies and missed violations. The lack of automated detection makes the system inefficient, especially in busy urban environments where continuous monitoring is required.

Human errors and limitations further reduce the reliability of the current system. Traffic personnel may experience fatigue, distraction, or reduced concentration during long hours of duty, leading to missed or incorrect identification of violations. In high-traffic situations, it becomes even more challenging to monitor multiple vehicles simultaneously. This affects the accuracy and consistency of traffic law enforcement.

The existing system also lacks an efficient method for identifying vehicles involved in violations. Number plates are not automatically captured or recognized, requiring manual recording or verification. This process is slow and prone to errors, making it difficult to track offenders or take appropriate action. Without proper identification, enforcing penalties becomes less effective.

These shortcomings clearly demonstrate the need for a more organized, centralized, and technology-driven system to maintain academic project work. A modern web-based platform must support structured submission, secure login, advanced search functions, and efficient browsing of past projects. It should also offer long-term, scalable digital storage to preserve documentation in a consistent format. Recognizing these gaps in existing practices forms the basis for developing Past Project Explorer, a unified digital repository designed to modernize the storage, management, and accessibility of academic projects.Another limitation of the existing system is its inability to provide centralized monitoring and integration across multiple locations. Traffic data collected from different cameras or checkpoints is often not connected, making it difficult for authorities to get a unified view of violations. This lack of integration reduces coordination between departments and delays decision-making. As a result, the system fails to support large-scale traffic management and smart city initiatives effectively

4.2 DRAWBACKS:

- Relies heavily on manual monitoring by traffic police
- Time-consuming and requires continuous human supervision
- High chances of human error and missed violations
- No real-time detection of helmet violations
- CCTV systems only record footage without automatic analysis
- Delay in identifying and taking action against violators
- No automatic number plate recognition system
- Inefficient handling of large traffic volumes
- Lack of proper data storage and analysis
- Not scalable for smart city or large-scale implementation

The existing traffic monitoring system has several limitations due to its dependence on manual supervision and basic surveillance tools. Traffic police are required to continuously observe riders to identify violations, which is both time-consuming and inefficient. In busy areas with heavy traffic, it becomes difficult to monitor every vehicle, leading to missed violations. Additionally, human factors such as fatigue, distraction, and reduced concentration can result in errors, making the system unreliable for consistent enforcement.

Another major drawback is the lack of automation and real-time processing in current systems. CCTV cameras only record footage without automatically detecting violations, requiring authorities to manually review videos, which delays action. There is also no efficient mechanism for automatic number plate recognition, making it difficult to identify and track offenders. Furthermore, the absence of proper data management and scalability limits the system's ability to handle large volumes of traffic data and support modern smart traffic solutions.

4.3 PROPOSED SYSTEM:

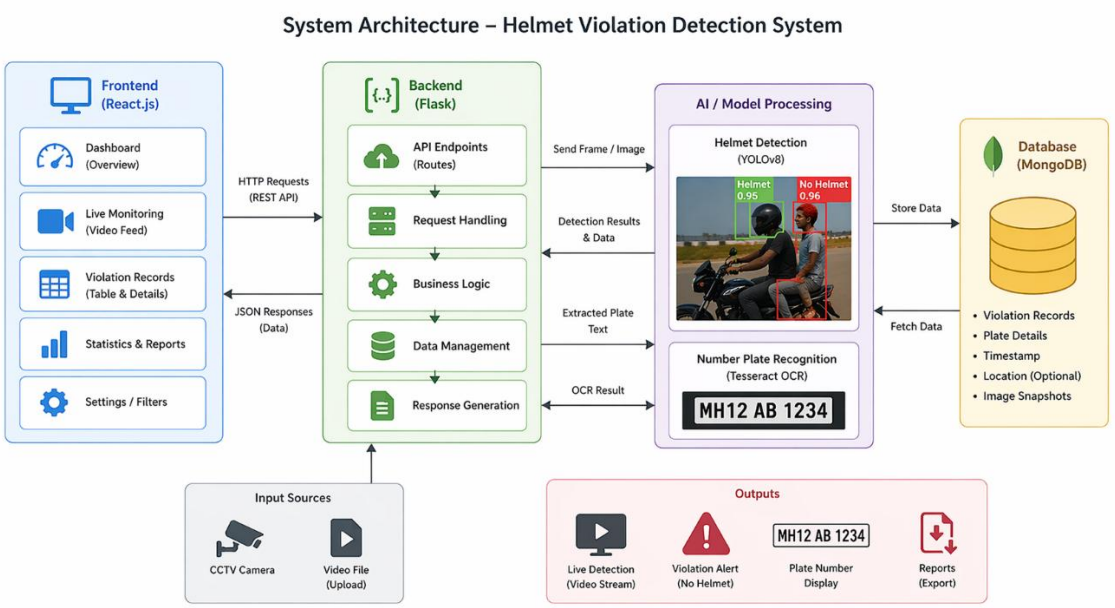


Fig 4.1 System Architecture of Traffic Violation Detection System

1. Project Submission and Initialization:

The project submission and initialization phase involves setting up the system and preparing it for execution. During this stage, the required software, libraries, and dependencies such as OpenCV, YOLO, and Tesseract OCR are installed and configured properly. The frontend and backend components are initialized, ensuring proper communication through APIs. Input sources such as video files or CCTV feeds are configured, and the system is tested to verify that all modules are functioning correctly. This phase ensures that the project environment is ready for smooth operation and real-time traffic violation detection.

2. Frontend Interface and Navigation:

The frontend interface of the system is designed to be simple and user-friendly, allowing easy navigation between different sections such as dashboard, live camera feed, and violation records. It provides clear visual elements and organized menus to help users quickly access information. The navigation system ensures smooth interaction and efficient monitoring of detected violations in real time.

3. Dynamic Project Listing:

Dynamic project listing refers to the ability of the system to display updated violation data automatically without requiring manual refresh. In this project, the frontend dashboard fetches data from the backend through APIs and presents it in a structured format such as tables or cards. Whenever a new helmet violation is detected, the system dynamically updates the list with details like number plate, timestamp, and captured image. This ensures that users always have access to the latest information.

Additionally, the dynamic listing feature improves user experience by allowing real-time interaction with the data. Users can easily browse through recent and past violations, and the system ensures quick loading and smooth updates. This functionality enhances the efficiency of monitoring and helps authorities take timely action based on the most recent data.

4. Authentication and Security:

Authentication is an essential component of the system that ensures only authorized users can access the dashboard and its features. The system can implement a secure login mechanism where users provide valid credentials before gaining access. This prevents unauthorized individuals from viewing or modifying sensitive data such as violation records and number plate details. By controlling user access, the system maintains confidentiality and ensures that only designated authorities can operate it.

Security measures are also applied to protect the data and communication within the system. Data exchanged between the frontend and backend is handled through secure API calls, reducing the risk of data breaches. Input validation and error handling help prevent malicious activities and ensure system stability. Additionally, secure storage practices and optional encryption techniques can be used to safeguard stored data, ensuring the reliability and integrity of the system.

5. Documentation Storage and Access:

Documentation plays an important role in ensuring that the system is easy to understand, use, and maintain. In this project, documentation includes details about system architecture, working methodology, technologies used, and instructions for installation and execution. Proper documentation helps developers understand the flow of the system and makes it easier to debug, update, or extend the project in the future. It also provides guidance to users on how to operate the dashboard and interpret the results effectively.

Access to the system is controlled to ensure that only authorized users can view and manage the data. The system can implement user roles and permissions, allowing different levels of access based on user responsibilities. For example, administrators may have full control, while other users may only have viewing rights. This controlled access improves security, prevents misuse, and ensures that sensitive information such as violation records and number plate details are handled responsibly.

6. Database Management:

Database management is important for storing and organizing the data generated by the system. In this project, details such as detected violations, number plate information, timestamps, and captured images can be stored in a structured format. This allows easy retrieval and management of records for monitoring and analysis.

The system ensures efficient data handling by supporting operations like storing, updating, and retrieving information when required. Proper database management helps maintain data accuracy and consistency, making it easier for authorities to track violations and generate reports.

7. Search and Filter Mechanism:

The search mechanism allows users to quickly find specific violation records by entering details such as number plate or date. This feature reduces the time required to manually browse through large amounts of data and helps in retrieving information efficiently.

The filter mechanism enables users to narrow down results based on criteria such as time, location, or type of violation. By displaying only relevant data, it improves clarity and makes monitoring and analysis more effective for the users.

8. API Integration and Communication:

API integration plays a crucial role in connecting the frontend and backend components of the system. In this project, APIs are used to send and receive data between the React-based interface and the Flask backend. When a violation is detected, the backend processes the data and sends the results through API endpoints, allowing the frontend to display updated information such as number plate details and violation status in real time.

Communication between components is handled efficiently using HTTP requests, ensuring smooth data flow throughout the system. Proper error handling and response management are implemented to maintain reliability and avoid failures during data transfer.

9. Deployment Setup:

Deployment setup involves preparing the system to run smoothly in a real environment. In this project, the backend server is configured using Flask, and the frontend is deployed using a development server or hosting platform. All required libraries and dependencies are installed to ensure proper execution of the system.

The system is tested after deployment to verify that all components are working correctly and communicating without errors. Proper configuration of ports, APIs, and input sources ensures smooth operation.

4.4 MERITS:

- Provides automated detection of helmet violations, reducing manual effort.
  Ensures continuous monitoring without human intervention.

- Improves accuracy in identifying violations using AI techniques.
  Minimizes human errors and missed detections.

- Enables real-time processing of traffic video data.
  Helps in immediate identification of rule violations.

- Automatically extracts number plate information using OCR.
  Simplifies identification of violating vehicles.

- Reduces workload on traffic police authorities.
  Allows them to focus on more critical tasks.

- Enhances road safety by enforcing helmet usage.
  Helps in reducing accident-related injuries.

- Enables quick search and filtering of violation records.
  Saves time in retrieving specific information.

- Scalable system suitable for smart city applications.
  Can handle large volumes of traffic data.

- Integrates multiple technologies like OpenCV, YOLO, and OCR.
  Demonstrates practical use of AI in real-world applications.

- Reduces dependency on manual CCTV monitoring.
  Improves efficiency of surveillance systems.

- Supports future enhancements and feature expansion.
  Can be upgraded for detecting more violations.

- Provides cost-effective solution for traffic management.
  Reduces need for additional manpower and resources.

4.5 ALGORITHM:

1. Input Acquisition:

Input acquisition is the first step in the system algorithm, where video data is obtained from sources such as CCTV cameras or pre-recorded traffic videos. The system initializes the video capture process using OpenCV, which allows it to read frames continuously from the input source. This step ensures that the system has a steady stream of visual data for further processing and analysis.

The acquired video frames are then prepared for subsequent stages of the algorithm. Each frame is extracted in sequence and passed to the processing pipeline for detection and analysis. Proper handling of input acquisition is important to maintain real-time performance and ensure that no frames are missed during processing. This step forms the foundation for accurate detection of helmet violations in the system.

2. Data Preprocessing:

Data preprocessing is an important step in the system that prepares the captured frames for accurate detection and analysis. Once the video frames are acquired, they are processed using OpenCV to improve image quality. This includes operations such as resizing the frames, converting them to grayscale, and reducing noise. These steps help in standardizing the input data and making it suitable for further processing by the detection model.

In addition, preprocessing techniques such as contrast enhancement and thresholding are applied to highlight important features in the image. These improvements increase the accuracy of both object detection and OCR processes. Proper preprocessing ensures that the system can effectively handle variations in lighting, background, and image quality, resulting in more reliable detection of helmet violations and number plate recognition.

3. Label Encoding and Data Balancing:

Data preprocessing is an important step in the system that prepares the captured frames for accurate detection and analysis. Once the video frames are acquired, they are processed using OpenCV to improve image quality. This includes operations such as resizing the frames, converting them to grayscale, and reducing noise. These steps help in standardizing the input data and making it suitable for further processing by the detection model.

In addition, preprocessing techniques such as contrast enhancement and thresholding are applied to highlight important features in the image. These improvements increase the accuracy of both object detection and OCR processes. Proper preprocessing ensures that the system can effectively handle variations in lighting, background, and image quality, resulting in more reliable detection of helmet violations and number plate recognition.

4. Indexing & Database Optimization:

Indexing is an important technique used to improve the speed and efficiency of data retrieval in the system. In this project, indexes can be created on key fields such as number plate, date, and time to allow quick searching of violation records. Instead of scanning the entire dataset, the system can directly access the required data using indexes, which significantly reduces response time. This is especially useful when handling large volumes of traffic violation data.

Database optimization further enhances system performance by organizing and managing data efficiently. Techniques such as proper data structuring, removing redundant data, and optimizing queries help in maintaining faster access and storage operations. Efficient database management ensures smooth handling of real-time updates and retrieval requests, making the system reliable and scalable for large-scale traffic monitoring applications.

5. Project Retrieval Logic:

Project retrieval logic refers to the method used by the system to fetch and display stored violation data when requested by the user. The system processes user queries such as search inputs or filter selections and retrieves the relevant records from the database through API calls. This ensures that users can quickly access specific information without delays.

The retrieval process is designed to be efficient and responsive, allowing real-time updates and smooth data loading on the dashboard. Proper handling of queries and responses ensures that accurate and relevant data is displayed. This improves user experience and supports effective monitoring of traffic violations.

6. Authentication Handling:

Authentication handling is a critical component of the system that ensures secure access to the application and its data. In this project, authentication can be implemented using a login mechanism where users are required to provide valid credentials such as a username and password before accessing the system. This process verifies the identity of the user and prevents unauthorized access to sensitive information like violation records and number plate details. Proper authentication handling helps maintain confidentiality and ensures that only authorized personnel, such as traffic authorities or administrators, can operate the system and view the data.

Session management techniques help maintain secure user sessions and automatically log out users after inactivity, reducing the risk of unauthorized usage. Additional security measures such as password encryption and secure communication protocols further strengthen the system, ensuring reliable and safe operation.

7. API Execution:

API execution is responsible for handling the communication between the frontend and backend components of the system. When the frontend requests data, such as detection results or violation records, the backend processes the request and sends an appropriate response through defined API endpoints. This allows the system to dynamically update information like number plate details and detected violations in real time.

The API execution process ensures smooth data flow by managing requests, responses, and error handling efficiently. It validates incoming data, processes it using backend logic, and returns structured results to the frontend. This enables seamless interaction between different modules and ensures that the system operates reliably and efficiently.

8. Deployment:

Deployment is the process of setting up the system so that it can run and be accessed in a real environment. In this project, the backend developed using Flask is configured to run on a local or remote server, while the frontend built with React is hosted using a development server or deployment platform. All required libraries, models, and dependencies are installed to ensure that the system functions correctly.

After deployment, the system is tested to verify that all components are working together without errors. Proper configuration of API endpoints, ports, and input sources ensures smooth communication between frontend and backend. This setup allows the system to perform real-time detection and provide reliable monitoring of helmet violations.

4.6 FLOWCHART:

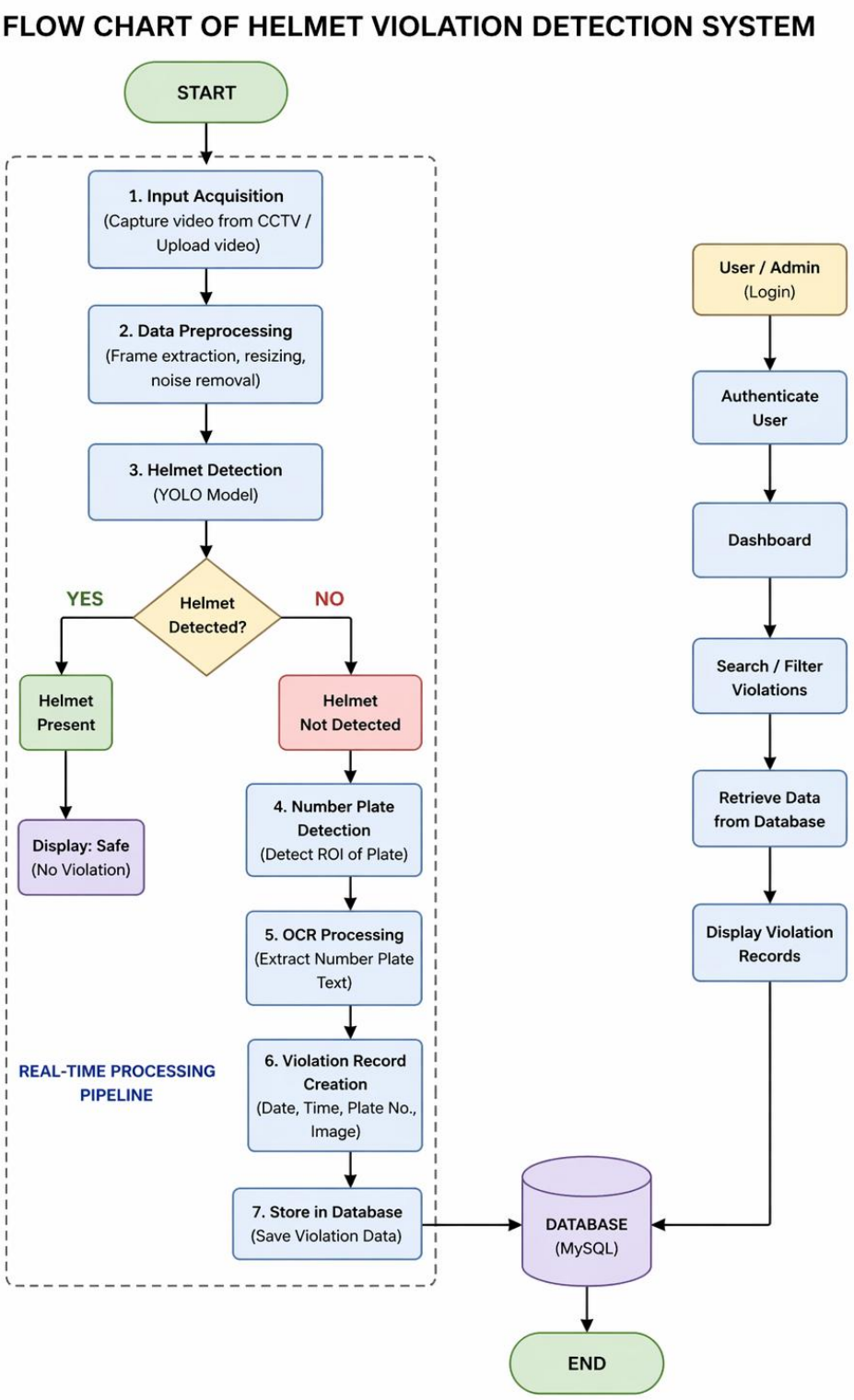


Fig 4.2 Flowchart of Traffic Violation Detection System

CHAPTER 5

RESULT AND DISCUSSION

5.1 RESULT:

The results of the proposed AI-Based Traffic Surveillance System demonstrate its effectiveness in detecting helmet violations among two-wheeler riders. The system was tested using video inputs from CCTV footage and recorded traffic videos, where it successfully identified motorcycles and analyzed helmet usage using a YOLO-based model. The detection process was able to highlight riders without helmets accurately and capture the corresponding frames for further processing. The integration of OpenCV ensured smooth frame extraction and real-time analysis, while the system maintained consistent performance across different lighting conditions and traffic densities. The detection accuracy was significantly improved compared to manual monitoring, and the system was able to process multiple frames efficiently without delays. These results confirm that the use of artificial intelligence and computer vision provides a reliable solution for automated traffic monitoring.

Furthermore, the system effectively performed number plate recognition using Tesseract OCR, enabling accurate identification of violating vehicles. The extracted number plate details, along with timestamps and captured images, were successfully displayed on the frontend dashboard in an organized manner. The API integration between the Flask backend and React frontend ensured seamless communication and real-time updates of data. Features such as search, filtering, and dynamic listing enhanced the usability of the system, allowing users to access and analyze violation records easily. Overall, the results indicate that the system is capable of reducing manual effort, improving detection accuracy, and providing an efficient platform for traffic law enforcement. The successful implementation of all modules demonstrates the practical applicability of system in real-world scenarios and its potential for future expansion in smart traffic management systems.

5.2 DISCUSSION:

The developed AI-based traffic surveillance system demonstrates a practical approach to automating helmet violation detection using computer vision and deep learning techniques. The integration of OpenCV, YOLO, and Tesseract OCR enables the system to process video data, detect violations, and extract number plate information efficiently. The results show that the system performs well under normal conditions, providing accurate detection and real-time monitoring capabilities. This highlights the potential of AI-based solutions in improving traffic law enforcement.

However, the system also has certain limitations that affect its performance in challenging environments. Factors such as poor lighting conditions, motion blur, and low-resolution video input can reduce detection accuracy. OCR performance may also be affected when number plates are unclear or partially visible. These challenges indicate the need for improved preprocessing techniques and more robust models to handle real-world variations effectively.

Despite these limitations, the system provides a strong foundation for future improvements and scalability. With enhancements such as better-trained models, integration of additional violation detection features, and cloud-based deployment, the system can be extended for large-scale applications. Overall, the project demonstrates the feasibility of using AI in traffic monitoring and opens opportunities for developing more advanced intelligent transportation systems

The system also highlights the importance of integrating multiple technologies to achieve an efficient and reliable solution. The coordination between frontend, backend, and AI models ensures smooth data flow and real-time updates, which are essential for effective monitoring. This integration demonstrates how combining different technologies can lead to a practical and scalable solution for modern traffic management.

CHAPTER 6

CONCLUSION AND FUTURE ENHANCEMENT

6.1 CONCLUSION:

The AI-Based Traffic Surveillance System developed in this project successfully demonstrates the use of computer vision and deep learning techniques to automate the detection of helmet violations among two-wheeler riders. By integrating OpenCV for video processing, a YOLO-based model for object detection, and Tesseract OCR for number plate recognition, the system is capable of analyzing traffic video data in real time and identifying violations with good accuracy. The system effectively captures frames of riders without helmets, extracts relevant information such as vehicle number plates, and presents the results through a user-friendly dashboard. This automation significantly reduces the need for manual monitoring and minimizes human error, making the system more reliable and efficient for traffic law enforcement.

In addition to improving detection efficiency, the project also highlights the importance of integrating different technologies to create a complete and functional system. The seamless communication between the React frontend and Flask backend ensures smooth data flow and real-time updates, while features such as search, filtering, and dynamic data display enhance usability. Although the current system focuses only on helmet detection, it provides a strong foundation for future enhancements such as detecting additional traffic violations, improving model accuracy, and implementing cloud-based solutions for large-scale deployment. Overall, the project demonstrates the potential of artificial intelligence in transforming traditional traffic monitoring systems into intelligent, automated solutions that contribute to improved road safety and smarter traffic management.

6.2 FUTURE ENHANCEMENT:

The proposed system can be significantly enhanced by expanding its capabilities beyond helmet detection to include multiple types of traffic violations. Future versions of the system can incorporate features such as seat belt detection for car drivers, red light violation detection, overspeed monitoring, and wrong lane detection using advanced deep learning models. By training more robust and diverse datasets, the system can improve its detection accuracy and adaptability to different real-world conditions. Additionally, integrating advanced object tracking algorithms can help monitor vehicle movement over time, enabling more precise violation detection. Enhancing the OCR module with more sophisticated techniques can also improve number plate recognition accuracy, especially in challenging conditions such as low lighting or motion blur.

Another important area for future enhancement is the scalability and deployment of the system in real-world environments. The system can be integrated with cloud-based platforms to enable centralized data storage, real-time monitoring, and large-scale deployment across multiple locations. Mobile application integration can allow authorities to access violation data remotely and take immediate action. Furthermore, implementing automated fine generation and notification systems can streamline the enforcement process. With the addition of AI-based analytics, the system can also provide insights into traffic patterns and help authorities make data-driven decisions. These enhancements will transform the project into a comprehensive intelligent traffic management system suitable for smart city applications.

APPENDIX

A. FRONTEND SOURCE CODE

A.1 Index Page

```
import { useState } from "react";
import axios from "axios";
export default function App() {
   const [file, setFile] = useState(null);
   const [result, setResult] = useState("");
   const upload = async () => {
   const data = new FormData();
   data.append("video", file);
   const res = await axios.post("http://localhost:5000/analyze", data);
   setResult(res.data.result);
  };
  return (
   <div style={{ textAlign: "center", marginTop: "50px" }}>
    <h2>Traffic Helmet Detection</h2>
    <input type="file" onChange={(e) => setFile(e.target.files[0])} />
    <br /><br />
    <button onClick={upload}>Upload</button>
    <h3>{result}</h3>
    </div>
  );
}
```

A.2 CSS Styling (style.css):

```
 body {
   margin: 0;
   font-family: Arial, sans-serif;
   background: #f4f6f8;
 }
 div {
   text-align: center;
 }h2 {
   color: #333;
 input {
   padding: 8px;
   margin-top: 20px;
```

```
}button {
  padding: 10px 20px;
  background: #007bff;
  color: white;
  border: none;
  border-radius: 6px;
  cursor: pointer;}
button:hover {
  background: #0056b3;}
h3 {margin-top: 20px;
  color: green;
  font-weight: bold;
}
```

A.3 Frontend Logic (script.js):

```
import { useState } from "react";
import axios from "axios";

export default function App() {
  const [file, setFile] = useState(null);
  const [result, setResult] = useState("");
  const upload = async () => {
    if (!file) return;
    const data = new FormData();
    data.append("video", file);
    const res = await axios.post("http://localhost:5000/analyze", data);
    setResult(res.data.result);
  };
  return (
    <div>
      <input type="file" onChange={(e) => setFile(e.target.files[0])} />
      <button onClick={upload}>Upload</button>
      <p>{result}</p>
    </div>
  );
}
```

B. BACKEND SOURCE CODE:

B.1 Server Setup (server.js):

```
from flask import Flask, request, jsonify
from flask_cors import CORS
import os
app = Flask(__name__)
CORS(app)
@app.route("/analyze", methods=["POST"])
def analyze():
file = request.files["video"]
path = os.path.join("uploads", file.filename)
os.makedirs("uploads", exist_ok=True)
file.save(path)
return jsonify({"result": "Helmet Detected"})
app.run()
```

B.2 MongoDB Schema (models/Project.js):

```
import mongoose from "mongoose";
mongoose.connect("mongodb://127.0.0.1:27017/trafficDB");
const violationSchema = new mongoose.Schema({
filename: String,
result: String,
createdAt: { type: Date, default: Date.now }
})
const Violation = mongoose.model("Violation", violationSchema);
export default Violation;
```

B.3 Authentication (jwt + bcrypt) :

```
const bcrypt = require("bcrypt");
const jwt = require("jsonwebtoken");
const User = require("./models/User");
app.post("/api/register", async (req, res)=>{
   const hashed = await bcrypt.hash(req.body.password, 10);
   await User.create({username:req.body.username, password:hashed});
   res.json({message:"User Registered"});
});
app.post("/api/login", async (req, res)=>{
   const user = await User.findOne({username:req.body.username});
   const valid = await bcrypt.compare(req.body.password, user.password);
```

```
    if(!valid) return res.status(401).json({error:"Invalid Credentials"});
    const token = jwt.sign({id:user._id}, "secretKey");
    res.json({token});
});
```

B.4 Protected Routes Middleware:

```
function verifyToken(req, res, next){
    const token = req.headers["authorization"];
    if(!token) return res.status(403).json({error:"No Token Provided"});
    jwt.verify(token, "secretKey", (err, decoded)=>{
        if(err) return res.status(401).json({error:"Invalid Token"});
        req.userId = decoded.id;
        next();
    });}
```

B.7 Search / Filter API:

```
app.get("/api/search", async (req, res)=>{
    let {keyword, year, tech} = req.query;
    let query = {};

    if(keyword) query.title = {$regex:keyword, $options:"i"};
    if(year) query.year = year;
    if(tech) query.technology = {$regex:tech, $options:"i"};

    const data = await Project.find(query);
    res.json(data);
});
```

C. SYSTEM FLOW SUMMARY:

1. Input Acquisition

- Captures video data from CCTV or recorded sources using OpenCV.

  Provides continuous frames for real-time processing and analysis.

2. Preprocessing

- Enhances image quality using resizing, grayscale conversion, and noise reduction. Improves accuracy of detection and OCR processes.

3. Storage:

- Stores violation data such as images, number plates, and timestamps. Ensures organized and efficient data management for future use.

4. Search & Retrieval:

- Allows users to quickly find specific records using search queries. Efficiently retrieves relevant data from the database.

5. Authentication:

- Ensures only authorized users can access the system. Protects sensitive data through secure login mechanisms.

6. Dynamic Rendering:

- Updates violation data on the dashboard in real time. Displays latest information without manual refresh.

7. Deployment:

- Sets up the system for execution in a real environment. Ensures smooth operation and communication between components.

OUTPUT

Main Interface (DASHBOARD) :



Fig 4.3 Dashboard

This page represents the main dashboard of the Smart AI Traffic Surveillance System, providing an overview of key traffic metrics. It displays real-time statistics such as total violations, average speed, peak traffic time, and safety score. Interactive charts like violation trends and capture analysis help in visualizing traffic patterns effectively. The sidebar navigation allows easy access to different modules such as live feed, violations, plate records, analytics, and settings.

Live Camera Feed :

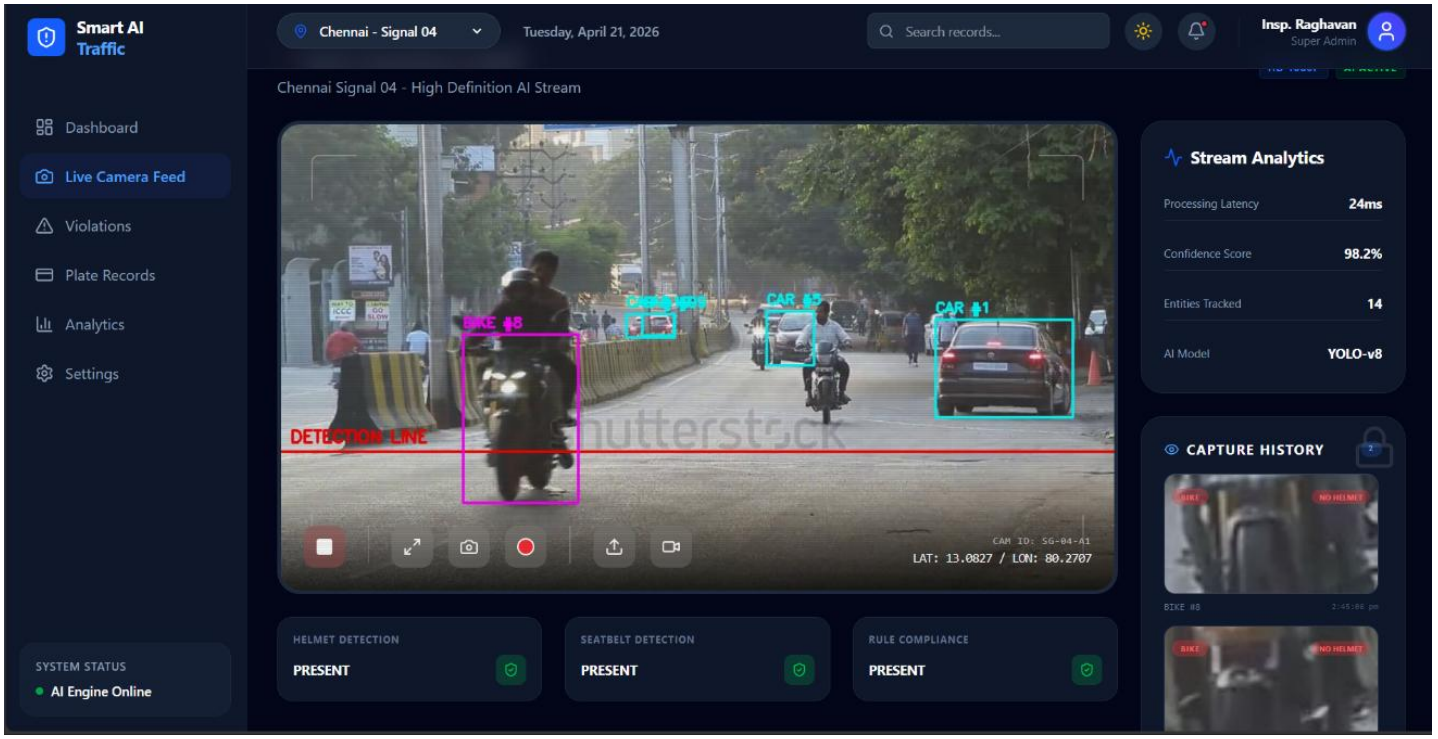


Fig 4.4 Live Camera Feed Page

This page displays the Live Camera Feed of the Smart AI Traffic Surveillance System, showing real-time video with object detection overlays. It highlights detected vehicles using bounding boxes and indicates their type, enabling continuous monitoring of traffic activity. This ensures accurate and efficient real-time detection performance.

Violation :

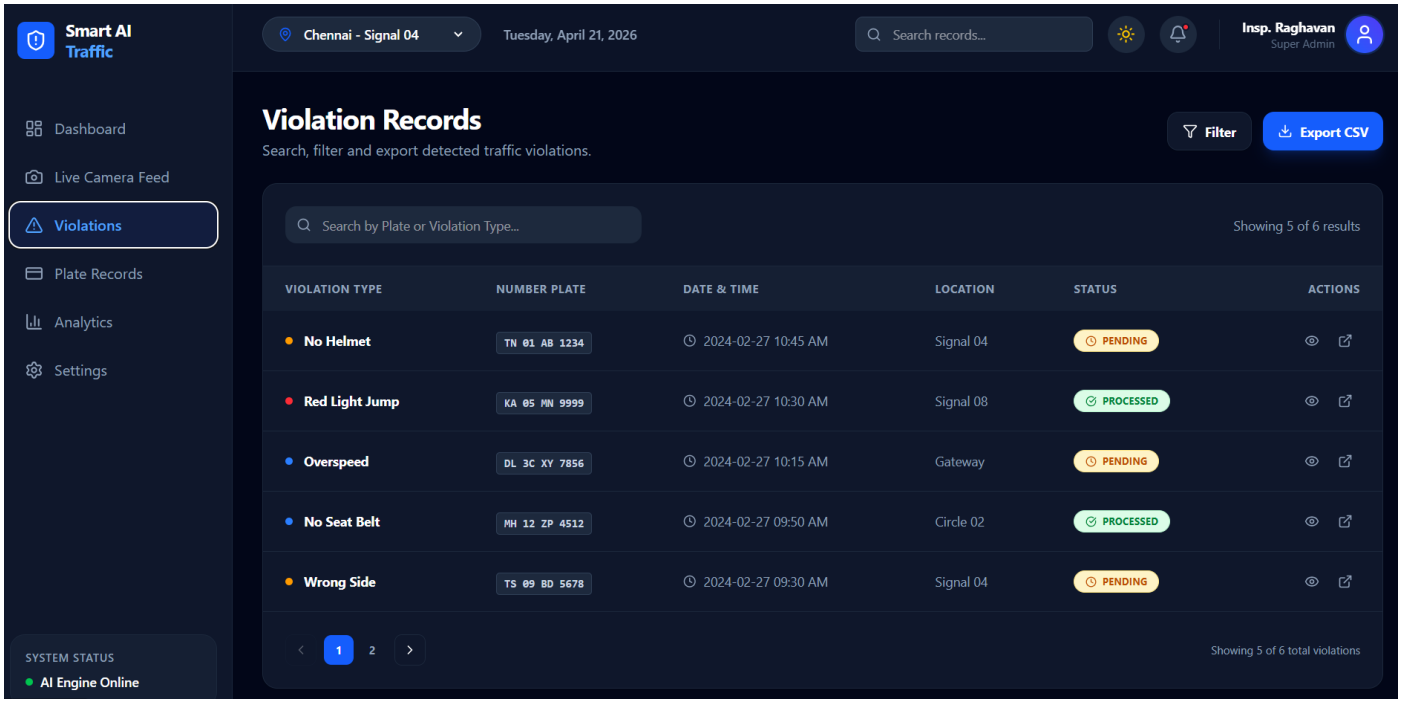


Fig 4.5 Violation Page

This page displays the Violation Records section of the system, where all detected traffic violations are listed in a structured table format. It includes details such as violation type, number plate, date and time, location, and processing status Users can easily search and filter records based on specific criteria like number plate or violation type, improving data accessibility. The export option allows downloading the records for reporting and analysis purposes. The interface also shows status indicators such as pending or processed, helping authorities track actions taken on each violation. Pagination and action buttons provide efficient navigation and detailed record management

Plate Records page:

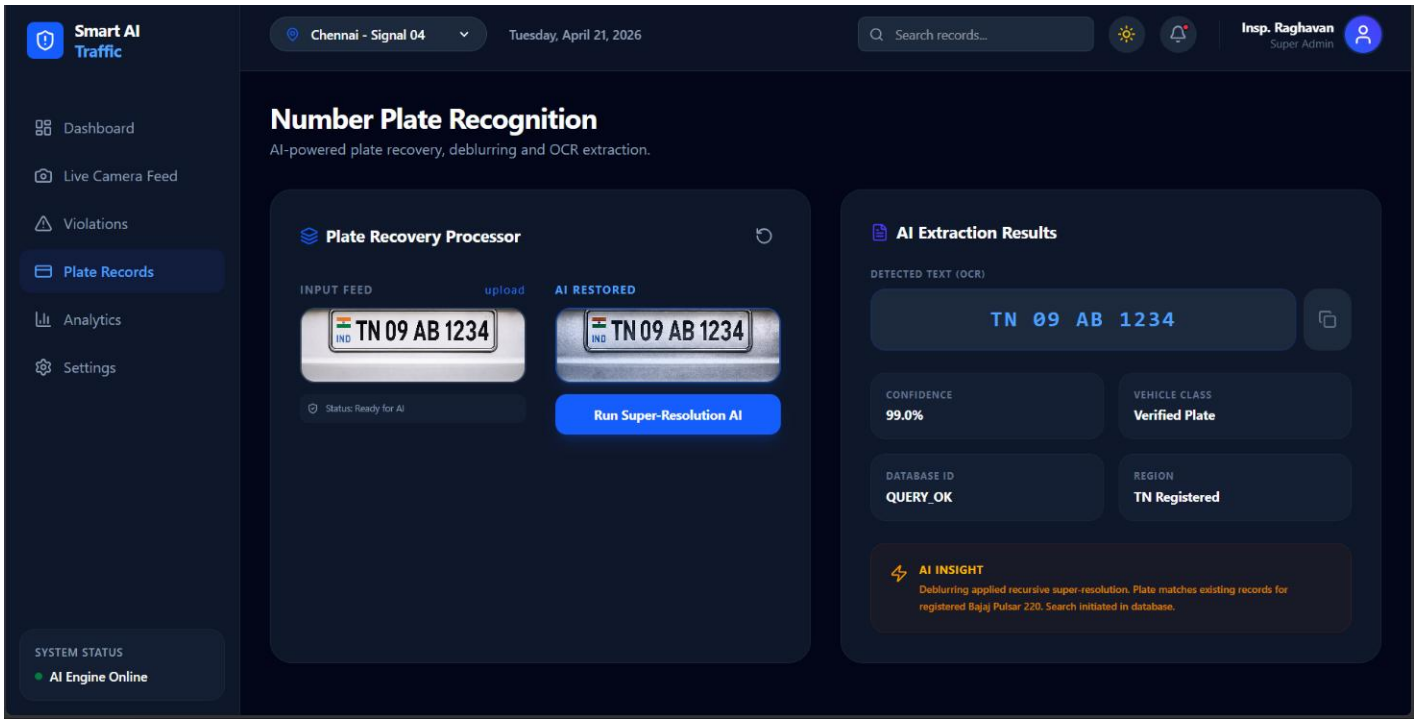


Fig 4.6 Plate Records Page

This page represents the Number Plate Recognition module, where AI is used to enhance and extract vehicle plate details. It shows both the input image and the AI-restored version for improved clarity. The system performs OCR to detect and display the number plate text along with confidence level and verification details. Additional information such as vehicle class, region, and database status is also presented

Analytics Page :



Fig 4.7 Analytics Page

This page represents the Number Plate Recognition module, where AI is used to enhance and extract vehicle plate details. It shows both the input image and the AI-restored version for improved clarity. The system performs OCR to detect and display the number plate text along with confidence level and verification details. Additional information such as vehicle class, region, and database status is also presented. The interface also provides AI insights and processing options like super-resolution, making it easier to recover blurred plates. This module helps in accurate identification of vehicles involved in traffic violations.

Settings Page :

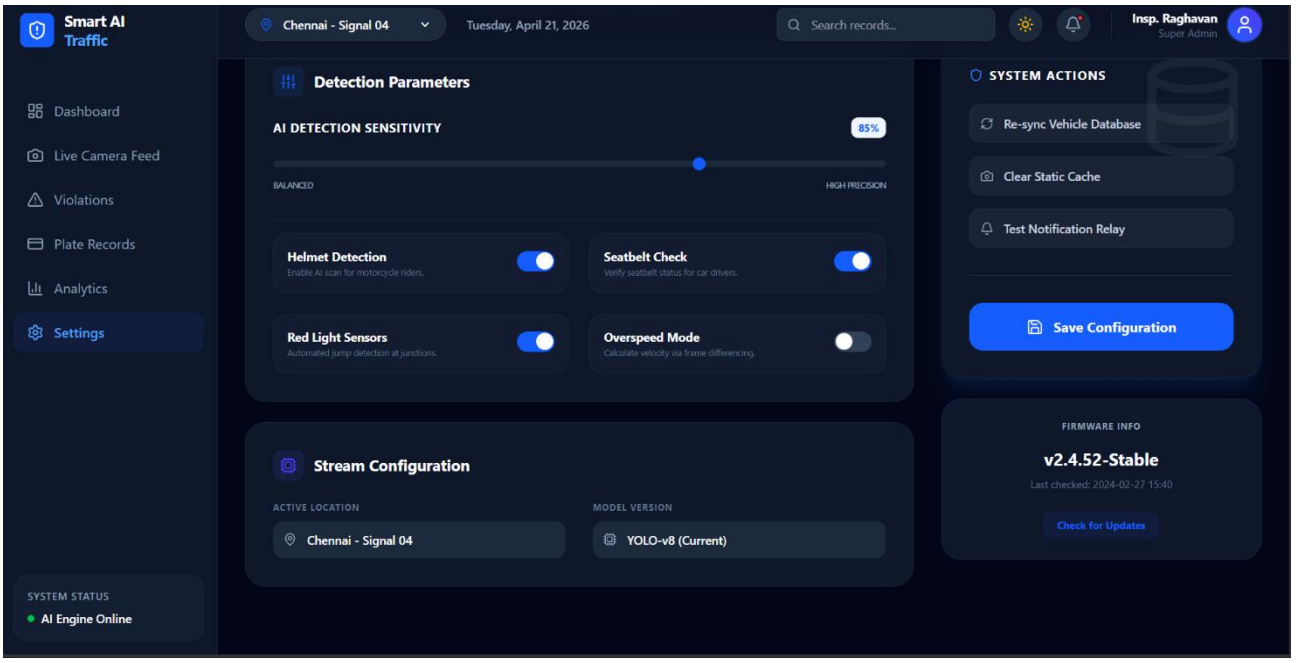


Fig 4.8 Setting page

This page represents the Settings module of the system, where users can configure detection parameters and system behavior. It allows adjustment of AI sensitivity and enables or disables features such as helmet detection, red light detection, and overspeed monitoring. The interface also includes system actions like database synchronization, cache clearing, and notification testing to maintain system performance. Users can save configurations to apply changes instantly. Additionally, stream configuration and firmware details are displayed, providing information about the active location and AI model version. This ensures better control, customization, and efficient management of the traffic surveillance system.

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